

Turn your iPad, iPod, iPhone into the smallest scope

Interesting innovations are happening to support oscilloscope functions on Apple devices working on iOS. A company named Oscium delivers innovative test equipment that leverage the Apple MFi interface. iMSO, an Oscium product, is the world's first mixed-signal oscilloscope (MSO) for iPod touch, iPhone and iPad. The company focuses on making test equipment more intuitive.

Based on Cypress's CY8CKIT-023 PSoC 3 device, iMSO is world's ultra-portable MSO with 2MHz bandwidth and up to 12MSa/s sample rate, ideal for hobbyists, students, and field sales and application engineers.

Innovative ASICs to address key challenges

Measuring instruments are the most complex real-time embedded systems which demand precision, scalability and speed. Vendors are now equally involved in highly innovative semiconductor R&D to come up with application-specific integrated circuits (ASICs) matching their special needs.

A key challenge in using a digital oscilloscope as a debug tool is its blind time. During this time, the user may miss critical signal events at his device under test. Typically, digital oscilloscopes today have a blind time ratio (the ratio of the active acquisition time to the blind time) of over 99.9 per cent.

Rao shares, "Some of the innovative ASICs demonstrate the ability to acquire, analyse and display the signal at a speed of greater than one million waveforms per second. The result is that the blind time between two acquisitions can go below 900 ns in standard operation mode and below 200 ns in segmentation acquisition mode." It also enables fast detection of rare signal anomalies without the need to make further setting on your scope.

True analogue bandwidth is another key criterion for an oscilloscope.



BitScope BS120U

"Bandwidth enhancement techniques like digital signal processing (DSP) boosting and frequency interleaving have many drawbacks like high noise and high total harmonic distortions," informs Bhatia. The invention of a new semiconductor called indium-phosphide (InP) and 3D Microcircuits Technology have proved to achieve a true analogue bandwidth as high as 32 GHz.

IBM's silicon germanium (SiGe) is another combination of elements which offers twice the performance of the previous generation along with real-time bandwidth beyond 30 GHz. Narasimhan informs, "The 130nm SiGe bipolar complementary metal-oxide semiconductor (BiCMOS) provides the next-generation, scalable, performance oscilloscope platform."

The performance level of SiGe is comparable to that of exotic materials like InP and gallium-arsenide (GaAs). High-speed bipolar transistors are present on the same die as the standard CMOS providing large-scale integration along with extremely high performance.

Future outlook

With continuously evolving technology, the key challenge for any test and measurement equipment is to keep up with the change. Bhatia shares, "With the buses becoming faster, chip-size becoming smaller and the quality of digital video becoming better, one

thing we should remember is that the expectation from the instrument used to test them is getting bigger!"

"It is difficult to predict what shape will an oscilloscope take in future. It may be a pocket-sized oscilloscope with bare minimum features like a

popular multimeter or Jules Verne's science-fiction oscilloscope. But one thing is sure that higher-bandwidth scopes with more information on rapidly changing waveforms will bring revolution in electronics due to better understanding and analysis of electronic circuits," says Kumar.

"Modularity will enable addition/removal of channels, upgradation of bandwidth and make the scopes sleeker, lighter and faster," informs Prabhanjana Rao, regional sales manager-South Asia, Lecroy.

"The oscilloscope would be a display device like a tablet PC with multiple inputs," adds Singh.

According to Baldua, "It will be smart and handy with big storage capabilities."

While USB has become the standard today, "In the future, Bluetooth connectivity will also help in quick transfer to mobile devices," says R.T. Swamy, head-T&M, Yokogawa.

"Oscilloscopes of the future will have tight integration with COTS technology to leverage heterogeneous computing architectures and inclusion of latest technologies like cloud computing," shares Subramanya. To deliver the ultimate user-experience, they will be highly customised modular equipment which can leverage a personal computer technology and connect to a network for controlled operation. ●

The author was a senior technology journalist at EYF